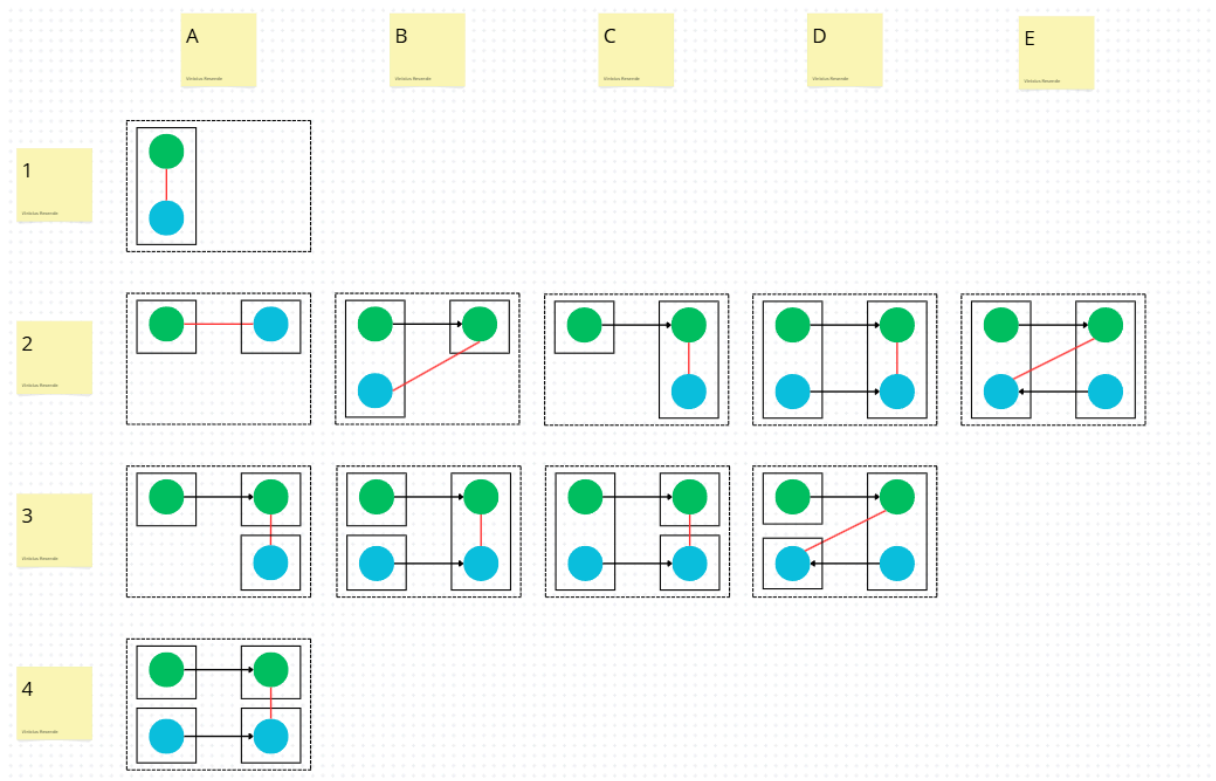


Data Flow (DF)



A1 - Variable definition in left, usage in right (in the same file)

```
src/main/java/org/example/Main.java [CHANGED]

2 2
3 3 public class Main {
4 4     public static void main(String[] args) {
5 -         Clue c = new Clue("lorem ipsum", -1);
5 +         Clue c = new Clue("lorem ipsum from previous", -1); // previous change
6 6         System.out.println("hello world");
7 +         boolean significance = c.isInsignificant(); // current change
7 8     }
8 9 }
```

On line 7, right (current) accesses a method of 'c', which was a variable modified by left (previous) on line 5.

A2 - Variable definition in right, usage in left (in different files)

```
☞ Main.java ×
1 public class Main {
2     public static int total = 100;
3
4     public static void main(String[] args) {
5         total += 50; // right change
6         Utils.printTotal();
7     }
8 }
```

```
☞ Utils.java ×
1 public class Utils {
2     public static void printTotal() {
3         System.out.println("Total atual: " + Main.total); // left change
4     }
5 }
```

right modifies the variable 'total' on line 5 of 'Main.java', left accesses it to print on line 3 of 'Utils.java'.

B2 - right has a method call from file 'A' to file 'B', in 'B' a variable is defined/modified. In file 'A', left accesses this variable.

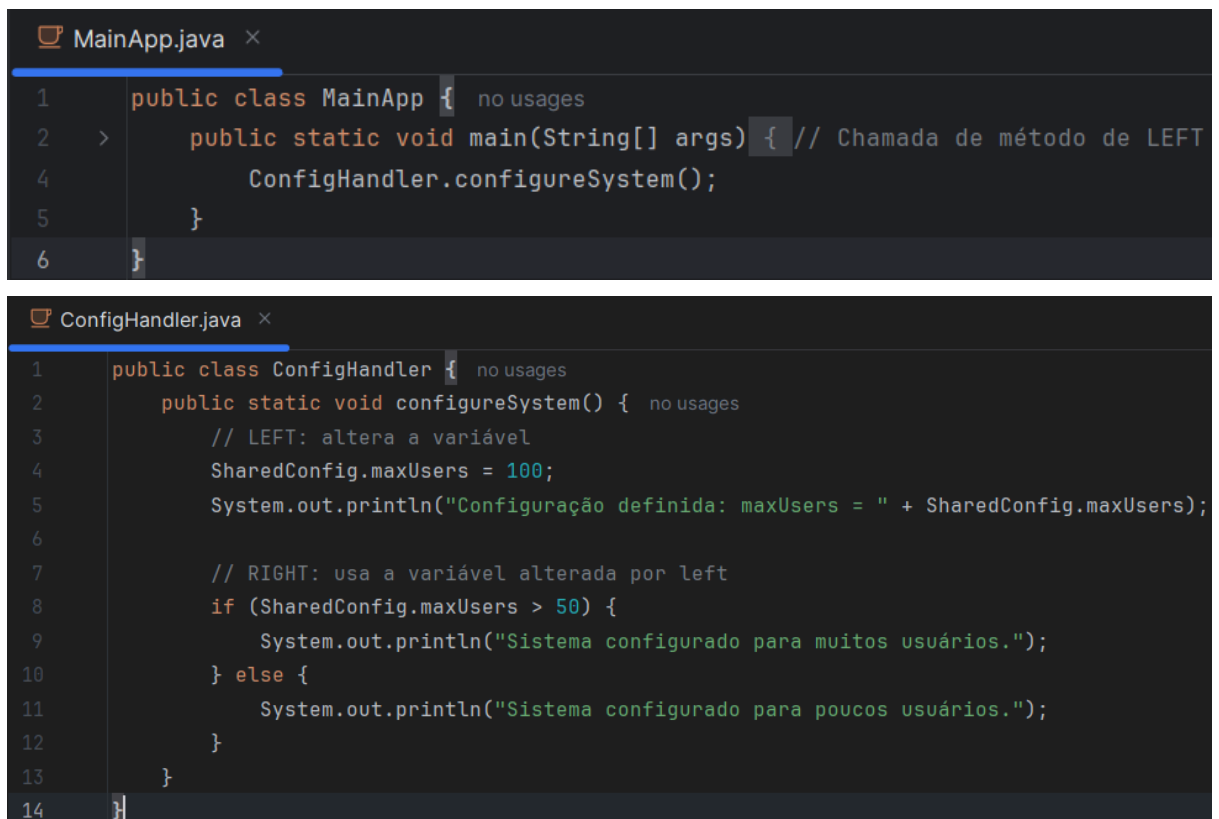
```
☞ Main.java ×
1 public class Main { no usages
2     public static void main(String[] args) { no usages
3         Config.atualizarConfig(); // RIGHT: chamada que altera a variável
4
5         int base = 10;
6         int resultado = base * Config.multiplier; // LEFT: usa a variável alterada
7         System.out.println("Resultado: " + resultado);
8     }
9 }
```

```
☞ Config.java ×
1 public class Config {
2     public static int multiplier = 2;
3
4     > public static void atualizarConfig() { multiplier = 5; // RIGHT: variável global modificada
7 }
```

In line 3 of 'MainApp.java', right adds a method call, which in 'ConfigHandler.java' modifies the variable 'multiplier' in line 2, used by left in line 6 of 'MainApp.java'.

Variable definition/modification is done using the right arrow key in 'ConfigHandler.java', while usage is done using the left arrow key in 'MainApp.java'.

C2 - Method call to a file where a variable is defined/modified by left; in that file, right uses that variable defined/modified by left.



```

MainApp.java x
1 public class MainApp { no usages
2   > public static void main(String[] args) { // Chamada de método de LEFT
3     ConfigHandler.configureSystem();
4   }
5 }
6 }

ConfigHandler.java x
1 public class ConfigHandler { no usages
2   public static void configureSystem() { no usages
3     // LEFT: altera a variável
4     SharedConfig.maxUsers = 100;
5     System.out.println("Configuração definida: maxUsers = " + SharedConfig.maxUsers);
6
7     // RIGHT: usa a variável alterada por left
8     if (SharedConfig.maxUsers > 50) {
9       System.out.println("Sistema configurado para muitos usuários.");
10    } else {
11      System.out.println("Sistema configurado para poucos usuários.");
12    }
13  }
14 }
```

In line 4 of the file 'MainApp.java', left calls the method 'configureSystem()' in the file 'ConfigHandler.java'. In line 4 of 'ConfigHandler.java', left modifies the variable 'maxUsers', which is used by right in line 8.

D2- In file 1 there is a method call to file 2, where `right` defines/modifies a variable. Additionally, there is another method call in file 1 to file 2, where `left` uses the same variable that `right` defined/modified.

```
src/main/java/org/example/Main.java CHANGED
▲
2 2
3 3 public class Main {
4 4     public static void main(String[] args) {
5 -         Clue c = new Clue("lorem ipsum", -1);
5 +         Clue c = new Clue("lorem ipsum from previous", -1); // previous change
6 6         System.out.println("hello world");
7 +         boolean significance = c.isInsignificant(); // current change
7 8     }
8 9 }
```

```
src/main/java/org/example/Clue.java CHANGED
▲
9 +     if (significance < 0) { // previous change
10 +         System.err.println("Normalizing negative significance " + significance + " to 0");
11 +     }
9 -     this.significance = significance;
12 +     this.significance = Math.max(significance, 0);
10 13 }
11 14
▼
▲
24 27 public boolean contains(String text) {
25 28     return description.contains(text);
26 29 }
30 +
31 + // current change
32 + /**
33 +  * Returns whether this {@link Clue} is insignificant, i.e., its significance is zero.
34 +  *
35 +  * @return Whether this {@link Clue} is insignificant
36 +  */
37 + public boolean isInsignificant() {
38 +     return significance <= 0;
39 + }
27 40 }
```

On line 5 in 'Main.java' there is a method call to 'Clue.java', where left modifies the variable 'significance' on line 12. On line 7 in 'Main.java' there is a call to the method 'IsInsignificant()' in 'Clue.java' which uses the variable 'significance' on line 38.

E2 - Left adds a method call in file 1 that modifies a variable in file 2. Later in file 2, Right adds a method call to file 1 that uses the variable previously modified by Left.

```
1 public class Main { no usages ykarocin *
2     public static void printCounter() { no usages new *
3         System.out.println("Counter value is: " + State.counter); //right change
4     }
5
6     public static void main(String[] args) { no usages ykarocin *
7         Worker.initialize(); // Left change
8     }
9 }
```

```
1 public class Worker { no usages ykarocin *
2     public static void initialize() { no usages new *
3         State.counter = 10; // left change
4         System.out.println("Counter Value: " + State.counter);
5         Main.printCounter(); // right change
6     }
```

On line 7 of “Main.java”, a method call was added that sets the value of the variable “counter” on line 3 of “Worker.java”. On line 5 of “Worker.java”, a method call was added that uses the value of the variable “counter” on line 3 of “Main.java”.

A3- In file 1, there is a method call to file 2, where right defines/modifies a variable. There is also a file 3, where left uses the variable defined/modified by right.

```
☞ Main.java ×
1 public class Main {
2     public static void main(String[] args) {
3         Config.inicializar();
4         Logger.logAmbiente();
5     }
6 }
```

```
Config.java ×
1 public class Config {
2     public static String ambiente = "dev";
3
4     public static void inicializar() {
5         ambiente = "prod"; // alteração de right
6     }
7 }
```

```
Logger.java ×
1 public class Logger {
2     public static void logAmbiente() {
3         System.out.println("Ambiente atual: " + Config.ambiente);
4     }
5 }
```

In line 3 of 'Main.java' there is a method call to the file 'Config.java', where the variable 'ambiente' is modified by right in line 5. In 'Main.java' there is also a call in line 4 to the file 'Logger.java', where left, in line 3, uses the variable 'ambiente'.

B3- In file 1 there is a method call to file 2, where `right` modifies/defines a variable. In file 3 there is also a method call to file 2, where `left` uses the variable modified by `right`.

```
ControllerA.java x
1 public class ControllerA {
2     public static void main(String[] args) {
3         SharedData.definirValor("Dados definidos por A");
4     }
5 }

SharedData.java x
1 public class SharedData { no usages
2     private static String valor; 2 usages
3
4     public static void definirValor(String novoValor) { no usages
5         valor = novoValor; // RIGHT
6     }
7
8     public static void imprimirValor() { no usages
9         System.out.println("Valor atual: " + valor); // LEFT
10    }
11 }

ControllerB.java x
1 public class ControllerB {
2     public static void main(String[] args) {
3         SharedData.imprimirValor();
4     }
5 }
```

In the file 'ControllerA.java' there is a method call on line 3 to the file 'SharedData.java' where right modifies the variable 'valor' on line 5. In 'ControllerB.java' there is a method call on line 3 to 'SharedData.java', where on line 9, left uses the variable 'valor' modified by right.

C3 - In file 1 there are two method calls, one to file 2, where `right` modifies/defines a variable, and another to file 3, where `left` uses the variable modified/defined by `right`.

```
☞ MainController.java ×  
1 public class MainController {  
2     public static void main(String[] args) {  
3         Writer.escreverValor("Texto definido por Writer");  
4         Reader.lerValor();  
5     }  
6 }
```

```
☞ Writter.java ×  
1 public class Writer {  
2     public static void escreverValor(String novoValor) {  
3         SharedData.valor = novoValor; //RIGHT|  
4     }  
5 }
```

```
☞ Render.java ×  
1 public class Reader {  
2     public static void lerValor() {  
3         System.out.println("Valor lido: " + SharedData.valor); //LEFT  
4     }  
5 }
```

In 'MainController.java' there are two method calls, one on line 3 to 'Writter.java' where right modifies the variable 'value' on line 3, and another on line 4 to 'Reader.java', where left uses the variable 'value' on line 3.

D3 - Left adds a method call in file 1 that sets the value of a variable in file 2. Right adds a method call in file 2 that uses that same variable in file 3.

```
1 public class Main { no usages ykarocin *
2     public static void main(String[] args) {
3         Worker.initialize(); // Left change
4     }
5 }
```

```
1 public class Worker { no usages ykarocin *
2     public static void initialize() { no usages new *
3         State.counter = 10; // left change
4         System.out.println("Counter Value: " + State.counter);
5         State.printCounter(); // right change
6     }
7 }
```

```
1 public class State { 1 usage new *
2     public static int counter = 0; 1 usage
3
4     public static void printCounter() { no usages new *
5         System.out.println("Counter value is: " + State.counter); //right change
6     }
7 }
```

In line 3 of “Main.java”, Left adds a method call that sets the value of “Counter” in line 3 of “Worker.java”. Right adds a method call in line 5 of “Worker.java” that uses the value of “counter” in line 5 of “State.java”.

A4 - In file 1 there is a method call to file 2, where `right` modifies/defines a variable. In file 3 there is a method call to file 4, where `left` uses the variable defined/modified by `right`.

```
MainApp.java x
1 public class MainApp {
2     public static void main(String[] args) {
3         ConfigWriter.defineConfig("Modo Escuro Ativado");
4     }
5 }
```

```
ConfigWriter.java ×
1 public class ConfigWriter {
2     public static void defineConfig(String valor) {
3         SharedConfig.setting = valor; // RIGHT
4         System.out.println("Config definida: " + valor);
5     }
6 }
```

```
SecondaryApp.java ×
1 public class SecondaryApp {
2     public static void main(String[] args) {
3         ConfigReader.usarConfig();
4     }
5 }
```

```
ConfigReader.java ×
1 public class ConfigReader {
2     public static void usarConfig() {
3         System.out.println("Config usada: " + SharedConfig.setting); // LEFT
4     }
5 }
```

In line 3 of 'Main.java', there is a method call to 'ConfigWriter.java', where on line 3, right modifies the variable 'setting'. In 'SecondaryApp.java', on line 3, there is a method call to 'ConfigReader.java', where left uses the variable 'setting' on line 3.